

Reference Points (First Two Letters)

Index	Meaning
B	Body frame origin (hub reference point or geometric center)
C	Spacecraft Center of Mass (CoM)
E	Earth center
S	Sun center
M	Moon center
N	Inertial frame origin
H	Hill frame origin
O	Generic reference origin

Frame Index (Last Letter)

Index	Meaning
N	Inertial frame (J2000 orientation)
B	Body frame
E	Earth-fixed frame (ECEF)
H	Hill / LVLH frame
S	Sun frame

Common Basilisk State Vectors

Variable	Meaning
r_CN_N	Position of spacecraft CoM relative to inertial origin, in inertial frame
v_CN_N	Velocity of spacecraft CoM relative to inertial origin, in inertial frame
r_BN_N	Position of body frame origin relative to inertial origin, in inertial frame
v_BN_N	Velocity of body frame origin relative to inertial origin, in inertial frame
r_CE_N	Position of spacecraft CoM relative to Earth center, in inertial frame
v_CE_N	Velocity of spacecraft CoM relative to Earth center, in inertial frame
r_CE_E	Position of spacecraft CoM relative to Earth center, in Earth-fixed frame
v_CE_E	Velocity of spacecraft CoM relative to Earth center, in Earth-fixed frame